Palmer Penguins Part 1: Exploratory Data Analysis and Simple Regression

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Figure 1: Penguins on an Antarctic shoreline, the starting point for a data-driven exploration of morphometric relationships.

Photo: Penguin colony. Licensed under CC BY 2.0 via Wikimedia Commons.

> **Note**
>
> **Palmer Penguins Series** – This is part 1 of the five-part modeling sequence; see the end of the post for the wider Palmer Penguins arc. Part 1 | Part 2 | Part 3 | Part 4 | Part 5

Introduction

How much can a single morphometric measurement reveal about an organism's body condition? The Palmer Penguins dataset provides an opportunity to test the claim that flipper length is a strong predictor of body mass in penguins. The exercise turns out to be one of the most instructive introductions to regression available.

The Palmer Penguins dataset, collected by Dr. Kristen Gorman at Palmer Station, Antarctica, contains morphometric measurements for three penguin species: Adelie (*Pygoscelis adeliae*), Chinstrap (*Pygoscelis antarcticus*), and Gentoo (*Pygoscelis papua*). Body mass is a key indicator of penguin health and reproductive success, and predicting it from easily-measured features has practical value for field researchers who may not always have access to a scale.

In this first post we walk through the complete exploratory data analysis pipeline, examine correlations among morphometric variables, and fit a simple linear regression model. The residual patterns that emerge set the stage for the multi-predictor models explored in Parts 2 through 5.

Motivations

The following considerations motivated this exploration:

- A hands-on project to practice exploratory data analysis on a well-documented ecological dataset rather than a toy example.
- Curiosity about whether a single measurement (flipper length) could meaningfully predict body mass, or whether the relationship is more complex.
- Simpson's Paradox appears in textbooks but seeing it arise organically in real data reinforces understanding; the Palmer Penguins dataset provides a clear illustration.
- A foundation for a multi-part series building from simple regression through random forests, with a clean EDA as the necessary first step.
- Practice in transparent model diagnostics: acknowledging limitations rather than presenting only favorable results.

Objectives

This post sets out to:

1. Conduct a structured exploratory data analysis on real ecological data, including summary statistics, distributions, and species-level comparisons.
2. Compute and interpret a correlation matrix to identify the strongest univariate predictor of penguin body mass.
3. Fit a simple linear regression model, extract coefficients and confidence intervals, and interpret R-squared in practical terms.
4. Evaluate model diagnostics—residual plots, outlier detection, and assumption checks—to identify where the model succeeds and where it falls short.

Errors and better approaches are welcome; see the Feedback section at the end.



Figure 2: Penguins in a library setting, symbolizing the research journey from data to understanding.

Prerequisites and Setup

The analysis pipeline that produces the derived data and figures for this post uses the following packages:

```
library(palmerpenguins)
library(tidyverse)
library(broom)
library(corrplot)
library(patchwork)
```

This document itself re-reads the pipeline's saved output rather than re-fitting anything, so rendering it needs only:

```
library(tidyverse)
library(knitr)
```

```r
theme_set(theme_minimal(base_size = 12))

penguin_colors <- c(
  "Adelie" = "#FF6B6B",
  "Chinstrap" = "#9B59B6",
  "Gentoo" = "#2E86AB"
)
```

Background note. This post assumes familiarity with basic R syntax and the tidyverse. No prior knowledge of regression modeling is required; the conceptual foundations are introduced below.

What is Exploratory Data Analysis?

Exploratory data analysis (EDA) is the practice of examining a dataset through summary statistics and visualizations before fitting any formal model. The purpose is to understand the structure of the data, detect anomalies, identify patterns, and generate hypotheses.

A useful analogy: EDA is to statistical modeling what a site survey is to architectural design. Before drawing blueprints, one needs to know the terrain. In our case, the "terrain" consists of morphometric measurements from 333 penguins across three species, and the patterns we uncover will guide every modeling decision in subsequent parts of this series.

Getting Started: Meeting the Penguins

Let us begin by examining the basic characteristics of the dataset.

```r
dataset_overview <- tibble::tribble(
  ~"Characteristic", ~"Value",
  "Total Observations",
    as.character(nrow(penguins_clean)),
  "Variables",
    as.character(ncol(penguins_clean)),
  "Species",
    as.character(
      dplyr::n_distinct(penguins_clean$species)
    ),
  "Islands",
    as.character(
      dplyr::n_distinct(penguins_clean$island)
    ),
  "Year Range",
    paste0(
      min(penguins_clean$year),
      "--",
      max(penguins_clean$year)
```

```
    )
)

kable(
  dataset_overview,
  col.names = c("", ""),
  caption = "Palmer Penguins Dataset Characteristics"
)
```

Table 1: Palmer Penguins Dataset Characteristics

Total Observations	333
Variables	8
Species	3
Islands	3
Year Range	2007–2009

The dataset includes 333 complete observations from three penguin species across three Antarctic islands, spanning 2007 to 2009. Each observation records bill length, bill depth, flipper length, body mass, species, island, sex, and year.

Species Composition

The three species are not equally represented. Understanding the balance matters because unequal sample sizes can affect statistical power in species-level comparisons.

```
kable(
  species_summary,
  caption = paste(
    "Species Distribution and",
    "Key Morphometrics"
  ),
  col.names = c(
    "Species", "N", "Body Mass (g)",
    "Flipper Length (mm)", "% of Dataset"
  )
)
```

Table 2: Species Distribution and Key Morphometrics

Species	N	Body Mass (g)	Flipper Length (mm)	% of Dataset
Adelie	146	3706	190.1	43.8
Chinstrap	68	3733	195.8	20.4
Gentoo	119	5092	217.2	35.7

Adelie penguins constitute the largest group (43.8%), followed by Gentoo (35.7%) and Chinstrap (20.4%). Gentoo penguins are notably heavier, with a mean body mass exceeding 5,000 g—roughly 1,400 g more than the other two species.

```
knitr::include_graphics("figures/eda-overview.png")
```

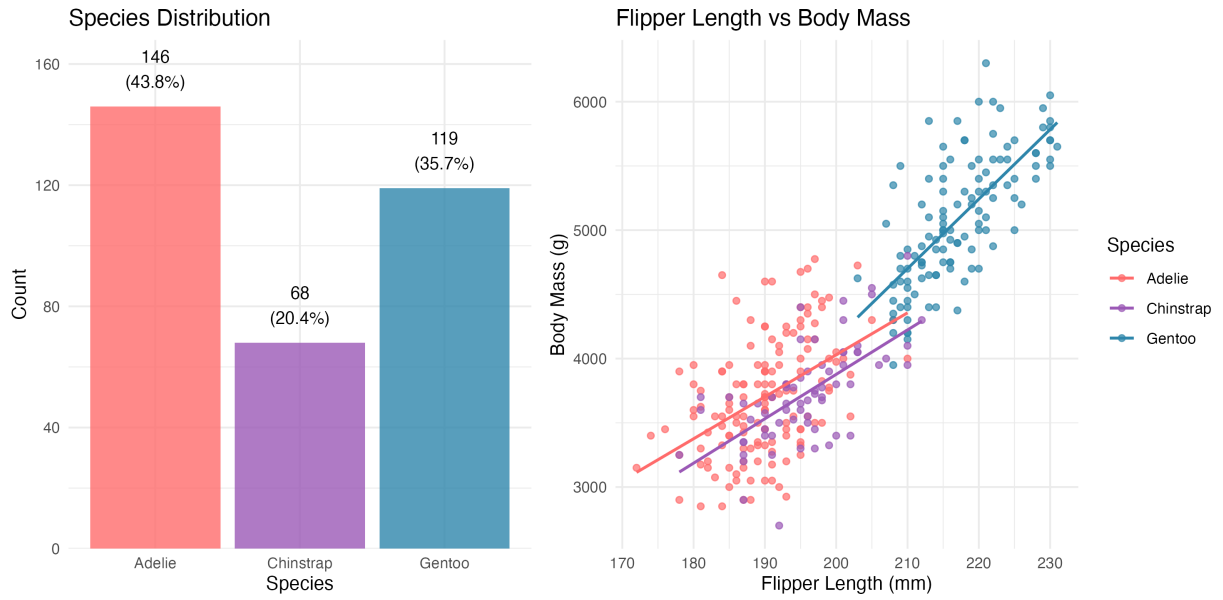


Figure 3: Species distribution and morphometric relationship overview. Left panel: sample sizes by species. Right panel: flipper length versus body mass with species-specific regression lines.

The right panel of the figure above reveals the central pattern: flipper length and body mass are strongly positively associated, but the relationship varies across species. Gentoo penguins cluster in the upper right, while Adelie and Chinstrap overlap substantially in the lower portion of the plot.



Figure 4: A group of penguins, each a unique data point waiting to be explored.

Species-Specific Morphometrics

A closer look at the per-species distributions reveals important nuances that a pooled analysis would obscure.

```
morphometric_summary <- penguins_clean |>
  group_by(species) |>
  summarise(
    n = n(),
    body_mass_mean = round(mean(body_mass_g), 0),
    body_mass_ci = round(
      1.96 * sd(body_mass_g) / sqrt(n()), 1
    ),
    flipper_length_mean = round(
      mean(flipper_length_mm), 1
    ),
    flipper_length_ci = round(
      1.96 * sd(flipper_length_mm) / sqrt(n()),
      1
    ),
    .groups = "drop"
```

```
  )

kable(
  morphometric_summary,
  caption = paste(
    "Morphometric Statistics by Species",
    "(+/- 95% CI)"
  ),
  col.names = c(
    "Species", "N", "Body Mass (g)",
    "+/- 95% CI",
    "Flipper Length (mm)", "+/- 95% CI"
  )
)
```

Table 3: Morphometric Statistics by Species (+/- 95% CI)

Species	N	Body Mass (g)	+/- 95% CI	Flipper Length (mm)	+/- 95% CI
Adelie	146	3706	74.4	190.1	1.1
Chinstrap	68	3733	91.4	195.8	1.7
Gentoo	119	5092	90.1	217.2	1.2

The interval around the Gentoo mean (approximately 5,092 g) is far removed from those for Adelie (3,706 g) and Chinstrap (3,733 g), whereas the Adelie and Chinstrap intervals overlap. Note that these intervals describe uncertainty about each species mean, not the spread of individual penguins, and comparing them by eye is a rough guide rather than a formal test; the underlying body mass distributions of Adelie and Chinstrap overlap more heavily still.

```
knitr::include_graphics(
  "figures/species-comparison.png"
)
```

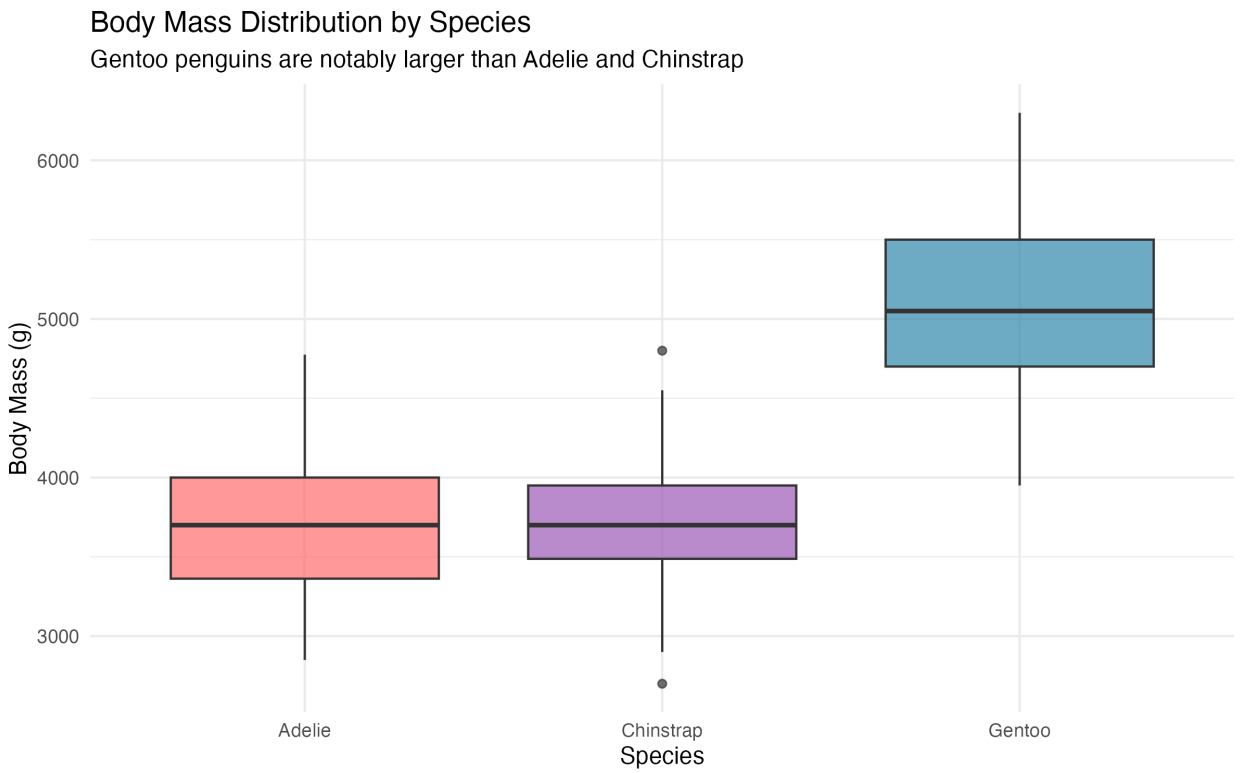


Figure 5: Body mass distributions by species. Gentoo penguins are substantially heavier, while Adelie and Chinstrap distributions overlap.

Correlation Analysis

Before fitting any model, we need to identify which morphometric variable is the strongest univariate predictor of body mass.

```
numeric_vars <- penguins_clean |>
  select(
    bill_length_mm, bill_depth_mm,
    flipper_length_mm, body_mass_g
  )
correlation_matrix <- cor(numeric_vars)
body_mass_cors <- correlation_matrix[
  "body_mass_g",
] |>
  sort(decreasing = TRUE)

correlation_summary <- tibble::tribble(
  ~"Variable",
  ~"Correlation with Body Mass",
  ~"Interpretation",
  "Flipper Length",
```

```
    round(body_mass_cors["flipper_length_mm"], 3),
    "Strongest predictor",
  "Bill Length",
    round(body_mass_cors["bill_length_mm"], 3),
    "Moderate positive",
  "Bill Depth",
    round(body_mass_cors["bill_depth_mm"], 3),
    "Moderate negative (pooled)"
)

kable(
  correlation_summary,
  caption = paste(
    "Morphometric Correlations",
    "with Body Mass"
  )
)
```

Table 4: Morphometric Correlations with Body Mass

Variable	Correlation with Body Mass	Interpretation
Flipper Length	0.873	Strongest predictor
Bill Length	0.589	Moderate positive
Bill Depth	-0.472	Moderate negative (pooled)

Flipper length dominates the correlation table with r = 0.873, well above bill length (r = 0.589) and bill depth (r = -0.472). The negative correlation between bill depth and body mass is a Simpson's Paradox artifact: within each species, the relationship reverses direction. We will return to this point in Part 2.

```
knitr::include_graphics(
  "figures/correlation-matrix.png"
)
```

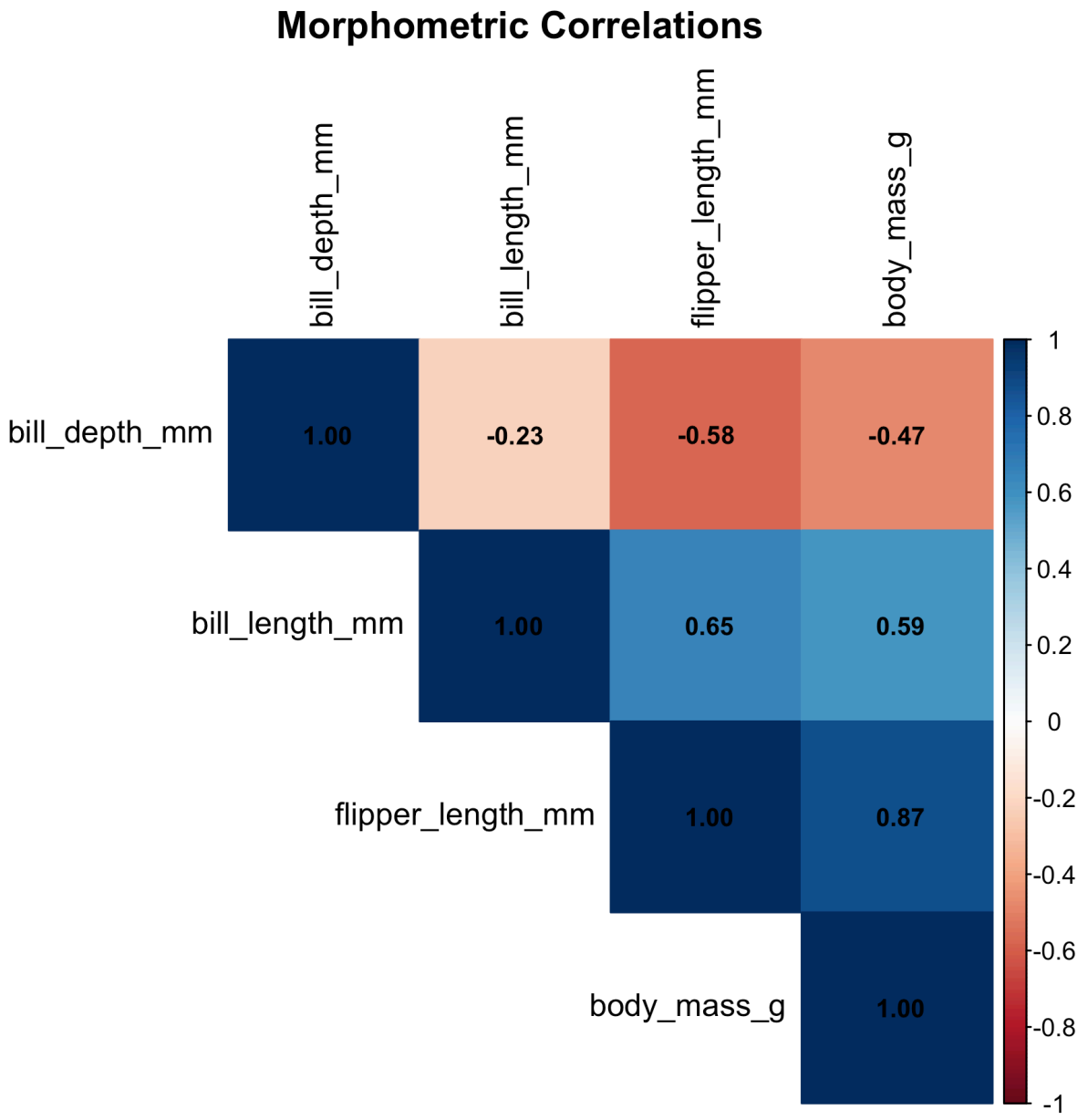


Figure 6: Correlation matrix of morphometric variables. Flipper length shows the strongest association with body mass (r = 0.87).

Building a Model: Simple Linear Regression

Given the strong bivariate correlation, flipper length is the natural choice for our first predictive model.

Fitting the Model

We fit an ordinary least squares regression of body mass on flipper length. The pre-computed coefficients and performance metrics are loaded from the analysis pipeline.

```
intercept <- model_coefficients$estimate[1]
slope <- model_coefficients$estimate[2]
slope_ci_lower <- model_coefficients$conf.low[2]
slope_ci_upper <- model_coefficients$conf.high[2]

model_summary <- tibble::tribble(
  ~"Metric", ~"Value", ~"Interpretation",
  "R-squared",
    sprintf("%.3f", model_metrics$r_squared),
    sprintf(
      "%.1f%% variance explained",
      model_metrics$r_squared * 100
    ),
  "RMSE",
    sprintf("%.1f g", model_metrics$rmse),
    "Typical prediction error",
  "Residual Std. Error",
    sprintf("%.1f g", model_metrics$residual_se),
    "Error scale adjusted for df",
  "F-statistic",
    sprintf("%.1f", model_metrics$f_statistic),
    sprintf(
      "p %s",
      format.pval(model_metrics$p_value, eps = 0.001)
    )
)

kable(
  model_summary,
  caption = "Simple Linear Model Performance"
)
```

Table 5: Simple Linear Model Performance

Metric	Value	Interpretation
R-squared	0.762	76.2% variance explained
RMSE	392.2 g	Typical prediction error
Residual Std. Error	393.3 g	Error scale adjusted for df
F-statistic	1060.3	p < 0.001

The model equation is:

Body Mass = -5872.1 + 50.2 x Flipper Length

The slope of 50.2 g/mm means that for each additional millimeter of flipper length, we expect body mass to increase by roughly 50 g on average. The 95% confidence interval for the slope is [47.1, 53.2] g/mm.

An R-squared of 0.762 indicates that flipper length alone accounts for approximately 76% of the variation in body mass—a strong result for a single predictor, but one that leaves meaningful residual variance unexplained.

Making Predictions

To illustrate the model in practical terms, here are predicted body masses for three representative flipper lengths:

```
example_flippers <- c(180, 200, 220)

prediction_ci <- predict(
  simple_model,
  newdata = data.frame(
    flipper_length_mm = example_flippers
  ),
  interval = "confidence"
)

predictions_display <- tibble::tibble(
  `Flipper Length (mm)` = example_flippers,
  `Predicted Body Mass (g)` = round(
    prediction_ci[, "fit"], 0
  ),
  `95% CI Lower` = round(prediction_ci[, "lwr"], 0),
  `95% CI Upper` = round(prediction_ci[, "upr"], 0)
)

kable(
  predictions_display,
  caption = paste(
    "Predicted Body Mass for",
    "Example Flipper Lengths"
  )
)
```

Table 6: Predicted Body Mass for Example Flipper Lengths

Flipper Length (mm)	Predicted Body Mass (g)	95% CI Lower	95% CI Upper
180	3155	3079	3232
200	4159	4116	4201

Flipper Length (mm)	Predicted Body Mass (g)	95% CI Lower	95% CI Upper
220	5162	5090	5233

A penguin with 200 mm flippers is predicted to weigh approximately 4,159 g, with a 95% confidence interval spanning only 85 g. The narrow intervals reflect the strong linear relationship, though they should be interpreted cautiously near the extremes of the observed flipper length range (172–231 mm).

```
knitr::include_graphics(
  "figures/simple-regression-model.png"
)
```



Figure 7: Simple linear regression of body mass on flipper length with 95% confidence band. Points are colored by species.

Checking Our Work: Model Diagnostics

A regression model is only as reliable as its assumptions. Before interpreting the results further, we must examine the residuals.

```r
outliers <- model_predictions |>
  filter(abs(standardized_residuals) > 2.5)

assumptions_check <- tibble::tribble(
  ~"Assumption", ~"Result", ~"Status",
  "Linearity",
    "Relationship appears approximately linear",
    "Met",
  "Independence",
    "Assumed from the sampling design; not testable here",
    "Assumed",
  "Normality",
    "Residuals approximately normal",
    "Reasonable",
  "Homoscedasticity",
    "Residual spread differs by species",
    "Violated",
  "Outliers",
    sprintf(
      "%d observations >2.5 SD", nrow(outliers)
    ),
    "Present",
  "Residual Std. Error",
    sprintf("%.1f grams", model_metrics$residual_se),
    "Acceptable"
)

kable(
  assumptions_check,
  caption = "Model Diagnostic Summary"
)
```

Table 7: Model Diagnostic Summary

Assumption	Result	Status
Linearity	Relationship appears approximately linear	Met
Independence	Assumed from the sampling design; not testable here	Assumed
Normality	Residuals approximately normal	Reasonable
Homoscedasticity	Residual spread differs by species	Violated
Outliers	5 observations >2.5 SD	Present
Residual Std. Error	393.3 grams	Acceptable

The most informative diagnostic is the residual plot below. The species separate along the horizontal axis simply because their flipper lengths differ, which is expected and not itself a fault; what matters is the vertical offset. The mean residual is -216 g for Chinstrap, +44 g for Adelie, and +69 g for Gentoo: the model over-predicts Chinstrap mass by a clear margin while under-predicting the other

two species more modestly. A model that ignores species therefore does not merely lose precision, it makes errors whose direction depends on which species the penguin belongs to. This is a clear signal that species membership carries information the model does not yet capture.

```
knitr::include_graphics(
  "figures/model-diagnostics.png"
)
```

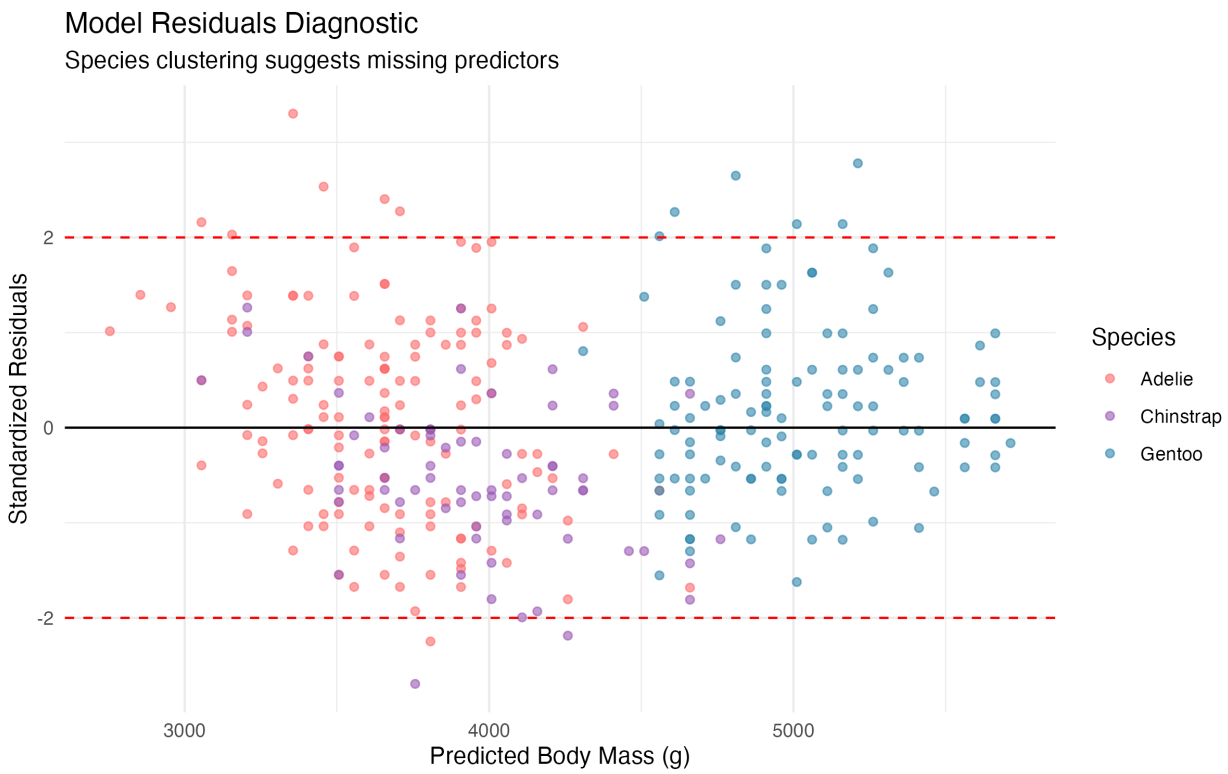


Figure 8: Standardized residuals versus predicted values, colored by species, with dashed lines at +/- 2 SD. Chinstrap residuals sit below zero more often than the other two species, the signature of an omitted categorical predictor.

Things to Watch Out For

1. **Simpson's Paradox.** The negative correlation between bill depth and body mass reverses within species. Always examine relationships both pooled and stratified before drawing conclusions.
2. **Residual clustering.** When residuals form visible groups, the model is missing a categorical predictor. In our case, species is the obvious candidate.
3. **Prediction extrapolation.** The model is fitted on flipper lengths between 172 and 231 mm. Predictions outside this range are unreliable and should be flagged as extrapolations.
4. **Confidence vs. prediction intervals.** The narrow confidence intervals in our predictions table describe uncertainty about the mean response, not about individual penguin masses. Prediction intervals would be substantially wider.

5. **Ecological confounding.** Body mass varies with sex, season, and breeding status—none of which are included in this model. Field researchers should account for these factors before using predictions for health assessments.



Figure 9: Penguins reflected in still water, a reminder to look beneath the surface of initial results.

What Did We Learn?

Lessons Learned

Conceptual Understanding

- Flipper length alone explains 76.2% of body mass variance (R-squared = 0.762), confirming it as the strongest single predictor among the available morphometric variables.
- The species-level clustering in residuals demonstrates why biological context must inform statistical modeling; ignoring species produces a model whose error has a species-dependent direction, over-predicting Chinstrap mass by roughly 200 g while under-predicting Adelie and Gentoo.
- Simpson's Paradox appears in the bill depth correlation: negative when pooled, positive within species. This is a textbook example that arises naturally in the data.
- Confidence intervals around the slope ([47.1, 53.2] g/mm) are narrow, reflecting a well-estimated relationship despite the model's structural limitations.

Technical Skills

- Loading pre-computed results from CSV files separates analysis from narrative, making the blog post faster to render and easier to maintain.
- Using `knitr::kable()` for all summary tables produces clean, consistent output across HTML and PDF formats.
- Extracting model coefficients and metrics from `broom::tidy()` and `broom::glance()` outputs (in the pipeline scripts) produces tidy data frames that integrate naturally into the reporting workflow.
- Setting a consistent color palette at the outset (`penguin_colors`) ensures visual coherence across all figures in the series.

Gotchas and Pitfalls

- Forgetting to remove incomplete cases before modeling can silently change sample sizes and produce misleading results. Always verify the observation count after data cleaning.
- Reporting R-squared without examining residual plots gives a false sense of model adequacy. A high R-squared does not guarantee that assumptions are met.
- Interpreting confidence intervals as prediction intervals overstates the model's precision for individual observations.
- Using `%>%` instead of `|>` in new code introduces an unnecessary dependency on magrittr. The native pipe is sufficient for all operations in this analysis.

Limitations

- **Temporal scope.** The data span only three years (2007–2009). Climate-driven changes in penguin morphology may have altered these relationships in the intervening years.
- **Geographic scope.** All observations come from the Palmer Station region. The model may not generalize to penguin populations in other Antarctic or sub-Antarctic locations.
- **Single predictor.** A univariate model cannot capture the multivariate biological reality of body mass determination. Sex, diet, and breeding status are all known to influence mass.
- **Missing variables.** The dataset does not include age, reproductive status, or feeding history—all of which are relevant covariates.
- **Measurement error.** Morphometric measurements have inherent imprecision (typically 1–2 mm for flipper length), which introduces attenuation bias into the slope estimate.
- **Species pooling.** Fitting a single regression line across three species conflates within-species and between-species variation, inflating the apparent predictive power of flipper length.

Opportunities for Improvement

1. **Add species as a predictor.** The residual clustering strongly suggests that a model including species (as explored in Part 2) will substantially improve fit.
2. **Include additional morphometric variables.** Bill length and bill depth may contribute explanatory power beyond what flipper length provides alone.
3. **Fit interaction terms.** The species-specific slopes visible in the EDA overview suggest that the flipper-mass relationship differs across species.

4. **Use cross-validation.** Splitting the data into training and test sets (as in Part 3) will provide a more honest estimate of predictive performance.
5. **Apply formal diagnostic tests.** The Breusch-Pagan test for heteroscedasticity and the Shapiro-Wilk test for normality would complement the visual diagnostics presented here.
6. **Compare with non-linear models.** Random forest and other flexible methods (Part 5) can capture non-linear relationships that OLS cannot.

Wrapping Up

This first post established the foundation for the Palmer Penguins analysis series. Starting with a thorough exploratory analysis, we identified flipper length as the dominant predictor of body mass and fitted a simple linear regression model that explains roughly three-quarters of the observed variation.

The most instructive finding was not the model's strength but its limitations. The species-level clustering in the residuals provided a clear, visual demonstration of why domain knowledge matters in statistical modeling. A high R-squared can coexist with systematic bias when an important categorical predictor is omitted.

For those undertaking a similar analysis, we recommend the following sequence: explore the data thoroughly before fitting any model, and always examine residual plots even when summary metrics look favorable. Resist the temptation to interpret a single model in isolation.

In conclusion, four points merit emphasis. First, flipper length is the strongest univariate predictor of body mass ($r = 0.873$, R-squared $= 0.762$), confirming that a single morphometric measurement carries substantial information about body condition. Second, the simple model has an RMSE of approximately 392 g, acceptable for broad field estimates but insufficient for precise individual predictions. Third, species-level residual clustering indicates that including species as a predictor will substantially improve the model, as confirmed in Part 2, where R-squared reaches 0.849. Fourth, Simpson's Paradox in the bill depth correlation underscores the importance of stratified analysis in ecological data and serves as a useful reminder that pooled associations can mislead.

> Preview: Part 2
>
> In Part 2, adding species information will improve the model's R-squared from 0.762 to 0.849—demonstrating why biological context matters in ecological modeling. Adding more morphometric measurements instead, without species, will turn out to achieve almost nothing.

See Also

Series Posts

- Part 2: Multiple Regression and Species Effects: Adding species as a predictor and fitting multiple regression.
- Part 3: Advanced Models and Cross-Validation: Model comparison via cross-validation.
- Part 4: Model Diagnostics and Interpretation: Comprehensive assumption checking.

- Part 5: Random Forest vs Linear Models: Comparing parametric and non-parametric approaches.

Key Resources

- Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLOS ONE*, 9(3), e90081.
- Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). *palmerpenguins: Palmer Archipelago (Antarctica) Penguin Data*. R package. https://allisonhorst.github.io/palmerpenguins/
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer. https://www.statlearning.com/
- Wickham, H., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly Media. https://r4ds.hadley.nz/

Reproducibility

This blog post is part of a **reproducible research compendium** built with ZZCOLLAB. All analysis code is separated from the narrative document.

Analysis Pipeline

The complete analysis consists of three reproducible scripts:

1. **01_prepare_data.R**: Load Palmer Penguins data, clean, and save derived data.
2. **02_fit_models.R**: Fit simple linear regression, extract coefficients and diagnostics.
3. **03_generate_figures.R**: Generate publication-quality figures from analysis results.

The Makefile declares each script's outputs as depending on the script and the prior stage's outputs, so `make docker-render-qmd` re-runs only the stages that are stale (or all three, on a fresh clone) before rendering – it does not just re-render against whatever derived data and figures happen to already be committed:

```
git clone <repository-url>
cd posts/pp-eda

make docker-build
make docker-render-qmd

open analysis/report/index.html
```

To run the pipeline (data, model, figures) without rendering – still only re-running stages that are stale:

```
make pipeline
```

Or run each script individually (outside the container; requires a local R environment matching renv.lock):

```
Rscript analysis/scripts/01_prepare_data.R
Rscript analysis/scripts/02_fit_models.R
Rscript analysis/scripts/03_generate_figures.R
quarto render analysis/report/index.qmd
```

Environment Information

```
env_info <- tibble::tribble(
  ~"Component", ~"Value",
  "R Version", R.version$version.string,
  "Platform", R.version$platform,
  "Analysis Date", as.character(Sys.Date())
)

kable(
  env_info,
  col.names = c("", ""),
  caption = "Analysis Environment"
)
```

Table 8: Analysis Environment

R Version	R version 4.6.0 (2026-04-24)
Platform	aarch64-apple-darwin23
Analysis Date	2026-11-09

Data Source

Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLOS ONE*, 9(3), e90081. Data accessible via the `palmerpenguins` R package: https://github.com/allisonhorst/palmerpenguins

Feedback

Feedback is welcome for:

- Errors or corrections to suggest

- Better approaches to any of these analyses
- Discussion of statistical methodology or ecological modeling
- Questions about the Palmer Penguins dataset or the ZZCOLLAB reproducibility framework
- General discussion of regression analysis for ecological data

Related posts in this cluster

This post is part of the *Palmer Penguins Analysis Arc* series. Recommended reading order:

0. **Precursor: zzcollab Analysis Profile Walkthrough** – set up the reproducible compendium used throughout this series
1. **Post 50: Palmer Penguins Part 1: Exploratory Data Analysis** (this post)
2. Post 51: Palmer Penguins Part 2: Multiple Regression
3. Post 52: Palmer Penguins Part 3: Cross-Validation
4. Post 53: Palmer Penguins Part 4: Model Diagnostics
5. Post 54: Palmer Penguins Part 5: Random Forest versus Linear
6. Post 55: Predictive Modeling of Penguin Body Mass
7. Post 56: Functional Plot Generation with purrr